

Claim Ledger

Claim Ledger for Chapter.docx

Working file: /Users/ra1/QuantumAccounting/BookChapter/sources/Chapter.original.docx

Extracted chapter text:

/Users/ra1/QuantumAccounting/BookChapter/extracted/Chapter.original.extracted.md

Sources Reviewed

- Quantum programming source: /Users/ra1/QuantumAccounting/BookChapter/sources/Quantum Programming January 19 2026.docx
- Odd-cycle source: /Users/ra1/QuantumAccounting/BookChapter/sources/Entanglement and Odd Cycle Game draft 02-12-2026.docx
- Information property source: /Users/ra1/QuantumAccounting/BookChapter/sources/Info-Property-Double-Entry-202503.docx
- Existing quantum-programming technical review:
/Users/ra1/QuantumAccounting/QuantumProgramming/quantum_programming_review.txt
- Existing odd-cycle review notes:
/Users/ra1/Library/CloudStorage/OneDrive-YaleUniversity/AFLS_Paper/LinFellingham/Review_Suggestions.md
- Information-property discussion notes in /Users/ra1/QuantumAccounting/InfoPropDoubleEntry
- External primary checks: Aharonov arXiv:quant-ph/0301040 and Shi arXiv:quant-ph/0205115
- External odd-cycle check: Drmota et al. arXiv:2406.08412

Status Key

- Supported: source result appears to justify the claim as written or with only ordinary citation support.
- Revise: underlying idea is usable, but current wording is too broad, imprecise, or underexplained.
- Verify: cited source was not in the working packet or the extracted files omit needed formal details.
- Remove or Recast: current claim is likely misleading unless substantially reframed.

Ledger

C01: The double-entry system remains relevant in the quantum era because its structure can repr

Chapter Location: Ch. lines 19-23

Claim: The double-entry system remains relevant in the quantum era because its structure can represent fundamental quantum operations.

Support Check: Supported by the quantum-programming source, but only as a representation or notation, not as physical quantum computation.

Status: Revise

Revision Action: Keep the thesis, but state that double-entry provides a classical representational language for quantum information structures.

C02: The advantage of quantum computing highlights advantages of double-entry over single-entry

Chapter Location: Ch. lines 23, 235

Claim: The advantage of quantum computing highlights advantages of double-entry over single-entry.

Support Check: The analogy is promising but not automatically entailed by the gate representation. It is supported mainly by the channel-capacity and odd-cycle models.

Status: Revise

Revision Action: Make the bridge explicit: "in stylized information-processing models, the relational structure of double entry can outperform scalar/single-entry representations."

C03: Decision-making, accounting numbers, and entropy are alternative representations of the sa

Chapter Location: Ch. lines 25, 37-65

Claim: Decision-making, accounting numbers, and entropy are alternative representations of the same activities.

Support Check: This cites Fellingham et al. 2022, which is not in the source packet. The chapter's equations were not extracted clearly enough to verify.

Status: Verify

Revision Action: Treat this as prior literature, not as something the chapter proves. Add a plain-English bridge before the equations.

C04: Kelly/log-growth/accounting equivalence supports the information-science framing.

Chapter Location: Ch. lines 43-55

Claim: Kelly/log-growth/accounting equivalence supports the information-science framing.

Support Check: Plausible, but the formal equations are missing from extraction and the underlying 2022 source is not included.

Status: Verify

Revision Action: Either provide a short derivation from the cited source or move detail to a footnote and keep the main text intuitive.

C05: The equivalence extends to quantum probabilities.

Chapter Location: Ch. line 65

Claim: The equivalence extends to quantum probabilities.

Support Check: Not verified from the packet. This is a high-risk bridge because quantum measurement is not just Bayesian updating.

Status: Verify

Revision Action: Use cautious wording: "the cited work argues..." and distinguish Bayesian conditionalization from quantum state update.

C06: Double-entry can process classical probability rules and may process quantum probability r

Chapter Location: Ch. lines 67-69

Claim: Double-entry can process classical probability rules and may process quantum probability rules.

Support Check: "May process" is safer than "can process." The quantum-programming source uses accounting notation plus external amplitude bookkeeping.

Status: Revise

Revision Action: Say the framework can encode selected quantum-information structures, with probabilities carried by separately tracked amplitudes.

C07: T-accounts encode qubit basis states and journal entries encode X, Z, S, and T-style opera

Chapter Location: Ch. lines 73-93

Claim: T-accounts encode qubit basis states and journal entries encode X, Z, S, and T-style operations.

Support Check: Supported by quantum-programming source lines 63-105.

Status: Supported

Revision Action: Retain, but add a roadmap table so nonexperts can track the mapping.

C08: Quantum gates are reversible unitary operations; journal entries preserve information anal

Chapter Location: Ch. lines 81-93

Claim: Quantum gates are reversible unitary operations; journal entries preserve information analogously.

Support Check: Reversibility claim is broadly correct for closed quantum systems, but reversibility is not unique to quantum computation.

Status: Revise

Revision Action: Avoid saying reversibility explains quantum superiority. Say it is one shared structural feature of unitary circuits and reversible accounting transformations.

C09: The Hadamard gate creates superposition; accounting notation stacks E and A rows.

Chapter Location: Ch. lines 97-105

Claim: The Hadamard gate creates superposition; accounting notation stacks E and A rows.

Support Check: Supported, but the accounting notation is incomplete unless amplitudes and normalization are specified.

Status: Revise

Revision Action: Add "the account notation records basis-state structure while amplitudes are recorded separately."

C10: Superposition allows information to be carried by two financial statements simultaneously.

Chapter Location: Ch. line 105

Claim: Superposition allows information to be carried by two financial statements simultaneously.

Support Check: This is an interpretive analogy, not a literal quantum statement about financial statements.

Status: Revise

Revision Action: Recast as: "the notation represents information across multiple linked statement rows, analogous to a superposition over basis states."

C11: Composite systems are represented by concatenated accounts or tensor products.

Chapter Location: Ch. lines 109-111

Claim: Composite systems are represented by concatenated accounts or tensor products.

Support Check: Must be stated carefully. In quantum mechanics, the state space of a composite system is the tensor product of state spaces; not every state factors as a tensor product.

Status: Revise

Revision Action: Correct the postulate language. Product basis states can be concatenated, but entangled states are superpositions of product states.

C12: CNOT transforms AEA into AAA when the first account controls the second.

Chapter Location: Ch. line 111

Claim: CNOT transforms AEA into AAA when the first account controls the second.

Support Check: This particular example is consistent with $E = 0$, $A = 1$, control = first, target = second.

Status: Supported

Revision Action: Retain. Avoid inheriting the reversed CNOT example flagged in the quantum-programming source review.

C13: Toffoli is universal for classical reversible computation and Toffoli plus Hadamard is universal for quantum computation.

Chapter Location: Ch. lines 117-119

Claim: Toffoli is universal for classical reversible computation and Toffoli plus Hadamard is universal for quantum computation.

Support Check: Supported by Aharonov arXiv:quant-ph/0301040 and Shi arXiv:quant-ph/0205115.

Status: Supported

Revision Action: Retain exact claim, but add "approximate arbitrary unitaries/circuits to arbitrary accuracy" if using the quantum universality claim.

C14: Since double-entry encodes H and Toffoli, it is universal for classical and quantum computation.

Chapter Location: Ch. line 119

Claim: Since double-entry encodes H and Toffoli, it is universal for classical and quantum computation.

Support Check: This is too strong as written. The source supports representational encoding of a universal gate set, not that the double-entry system itself is a computational device.

Status: Revise

Revision Action: Say "the framework can represent a universal gate set" and "therefore has the representational vocabulary needed to describe universal quantum computation."

C15: Symmetry, balance, and reversibility explain the capacity.

Chapter Location: Ch. line 121

Claim: Symmetry, balance, and reversibility explain the capacity.

Support Check: Plausible conjecture, not a theorem in the sources.

Status: Revise

Revision Action: Label as conjecture or interpretive explanation. Tie channel-capacity advantage more directly to articulation and relational structure.

C16: The superiority of quantum computers over classical computers arises largely from entanglement.

Chapter Location: Ch. line 123

Claim: The superiority of quantum computers over classical computers arises largely from entanglement.

Support Check: Overbroad. Entanglement is a key resource in many quantum advantages, but not the sole or universal explanation.

Status: Remove or Recast

Revision Action: Use "Entanglement is one defining resource in many quantum information tasks."

C17: Bell states can be generated by H followed by CNOT and represent nonseparable account relationships

Chapter Location: Ch. lines 129-139

Claim: Bell states can be generated by H followed by CNOT and represent nonseparable account relationships.

Support Check: Supported by the odd-cycle source and standard quantum theory.

Status: Supported

Revision Action: Retain, but explain "nonseparable" before "nonlocal" for nonexpert readers.

C18: Entanglement gives two-bit channel capacity by allowing four journal-entry messages to be distinguished through operations on one account.

Chapter Location: Ch. lines 145-185

Claim: Entanglement gives two-bit channel capacity by allowing four journal-entry messages to be distinguished through operations on one account.

Support Check: Supported by the information-property source, with a qualification: the original paper frames the mechanism as superposition plus articulation; "Bell state" language is analogical and appears in Appendix B.

Status: Revise

Revision Action: State the accounting conditions precisely: two articulating statement rows and at least two accounts satisfying the articulation identities.

C19: n-bit channel capacity is achieved with n entangled accounts.

Chapter Location: Ch. line 185

Claim: n-bit channel capacity is achieved with n entangled accounts.

Support Check: The information-property source supports n-bit capacity under specific n-account/two-statement articulation conditions.

Status: Revise

Revision Action: Replace with: "in the n-account extension, the model produces n bits when all n accounts satisfy the articulation conditions."

C20: Classical 3-seat odd-cycle maximum is 5/6; entangled case reaches $1/2 + \sqrt{3}/4$; superposition case reaches $1/2 + \sqrt{2}/4$

Chapter Location: Ch. lines 189-203

Claim: Classical 3-seat odd-cycle maximum is $5/6$; entangled case reaches $1/2 + \sqrt{3}/4$; superposition without entanglement reaches $1/2$.

Support Check: Supported by the odd-cycle source. The exact quantum value equals $\cos^2(\pi/12)$. Drmota et al. gives the general odd-cycle quantum limit $\cos^2[\pi/(4n)]$.

Status: Supported

Revision Action: Use exact value first, approximation second. Add a simple nontechnical explanation of why $5/6$ is the classical ceiling.

C21: Entanglement is the source of coordination advantage; superposition without entanglement c

Chapter Location: Ch. lines 205-207

Claim: Entanglement is the source of coordination advantage; superposition without entanglement can be detrimental.

Support Check: Supported within the specific three-case odd-cycle model.

Status: Revise

Revision Action: Add "in this model" and avoid generalizing to all quantum strategies or all organizational settings.

C22: The classical odd-cycle game is analogous to single-entry, while the quantum version is an

Chapter Location: Ch. line 209

Claim: The classical odd-cycle game is analogous to single-entry, while the quantum version is analogous to double-entry.

Support Check: This is interpretive, not a theorem. It can be valuable if framed as an analogy.

Status: Revise

Revision Action: Say "suggests an analogy" and clarify that double-entry remains a classical accounting architecture.

C23: In agency settings, entanglement is a productive resource and group measurement can domina

Chapter Location: Ch. lines 213-227

Claim: In agency settings, entanglement is a productive resource and group measurement can dominate individual measurement under synergy.

Support Check: The Demski et al. 2008 source is cited but not included in the working packet. The chapter maps synergy to entanglement, which is interpretive.

Status: Verify

Revision Action: Keep the Demski result only if the original paper is checked. Soften "entanglement contributes directly to production" to "relational measurement preserves information about joint production."

C24: More severe agency problems expand the range over which group measurement dominates.

Chapter Location: Ch. lines 225-227

Claim: More severe agency problems expand the range over which group measurement dominates.

Support Check: Not verified from the packet.

Status: Verify

Revision Action: Check Demski et al. 2008 before making this a major conclusion.

C25: The chapter shows double-entry supports vector processing, superposition, and entanglement

Chapter Location: Ch. lines 231-237

Claim: The chapter shows double-entry supports vector processing, superposition, and entanglement.

Support Check: The chapter can show formal analogues, not literal physical quantum properties.

Status: Revise

Revision Action: Use "admits a vector representation," "has a structural analogue of superposition," and "can model nonseparable relational structures."

C26: Double-entry processes information in a way that cannot be achieved classically.

Chapter Location: Ch. lines 233-235

Claim: Double-entry processes information in a way that cannot be achieved classically.

Support Check: Misleading because double-entry is classical. The intended comparison is against scalar/single-entry representations under specific models.

Status: Remove or Recast

Revision Action: Say "cannot be achieved by the corresponding scalar/single-entry representation in these models."

C27: The analysis explains the durability of double-entry over five centuries.

Chapter Location: Ch. line 235

Claim: The analysis explains the durability of double-entry over five centuries.

Support Check: Too strong as a historical causal claim. The result offers one information-theoretic explanation, not a complete historical explanation.

Status: Revise

Revision Action: Use "offers one reason" or "helps explain."

C28: The framework has pedagogical value for researchers, educators, and practitioners.

Chapter Location: Ch. line 237

Claim: The framework has pedagogical value for researchers, educators, and practitioners.

Support Check: Supported as a reasonable conclusion if overstatements are controlled.

Status: Supported

Revision Action: Retain. It may become a stronger closing than the broad "quantum era" claim.

Highest-Priority Corrections

1. Change the main claim from "double-entry is universal for quantum computation" to "double-entry can represent a universal quantum gate set when paired with explicit amplitude bookkeeping."
2. Replace literal-sounding claims of accounting entanglement/superposition with "structural analogues" unless a formal Bell-state model is actually being used.
3. Add exact conditions to the channel-capacity result: two statement rows, at least two accounts, and articulation identities.
4. Treat Section 1 and Section 3.4 as citation-dependent until Fellingham et al. 2022 and Demski et al. 2008 are checked.
5. Reframe the conclusion around "articulation as code": information lives in relationships among accounts/statements, not merely in individual balances.